Stat 992-001: Statistical methods for QTL mapping (Spring, 2011)

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Web	`http://www.biostat.wisc.edu/~kbroman`
Office hours	W 9:00–10:00am, Th 2:00–3:00pm, or by appointment
Lectures	MW 2:25–3:40pm, 5295 Medical Sciences Center
Textbooks	Broman and Sen, *A guide to QTL mapping with R/qtl* [required]
	Dalgaard, *Introductory statistics with R*, 2nd edition [recommended]
	Venables and Ripley, *Modern applied statistics with S*, 4th edition [recommended]
Course web	`http://www.biostat.wisc.edu/~kbroman/teaching/qtltopics`

Objectives

QTL are quantitative trait loci: genetic loci that contribute to variation in a quantitative trait (such as body weight or blood pressure). In this course, we will give an overview of statistical methods for mapping QTL in experimental crosses.

The course will focus largely on mouse genetics; however, it should also be valuable to students interested in human genetics or in animal and plant breeding. Mouse crosses are simple, and discussion of QTL mapping in this context can allow the basic principles to be elucidated with greater clarity.

The first half of the course will focus on basic aspects of QTL mapping, including interval mapping, significance thresholds, the incorporation of covariates, multiple QTL models, and the use of the R/qtl software for these analyses. In the second half of the course, we will focus on more complex cross designs as well as eQTL mapping.

The course will be aimed at PhD students in Statistics or Biostatistics, though we also welcome graduate students from the biological sciences. The statistical aspects should be accessible to any student with knowledge of analysis of variance and linear regression.

Prerequisites

Genetics 466 or equivalent; one of Statistics 310, 571, 610, or 710, or equivalent (that is, some basic statistics course); some knowledge of R.

Computing

We will use R (`http://www.R-project.org`) and R/qtl (`http://www.rqtl.org`) throughout the course.

Homework

There will be 4 homework assignments, which will mostly involve computational analyses to explore different aspects of QTL mapping.

Final project

There will be no exams in the course. Rather, each student will complete a final project: a 6–10 page paper describing the analysis of a QTL mapping dataset. This will be along the lines of a regular scientific paper: Introduction, Methods, Results, Discussion. Computer code is to be placed in an Appendix.

Academic honesty

You are *encouraged* to discuss homework assignments and your final project with other students. However, you must not present others' work as your own. If you work with other students in solving problems, make sure that you each write up your own solutions independently. It is not acceptable for one student to write a solution for another student to copy.

Grading

Your grade will be determined by homework assignments (70%) and the final project (30%). I will assign course grades according to the following scale, though this may be adjusted if it turns out that I've been grading too harshly.

Percentage	Grade
94 – 100%	A
87 – 94%	AB
80 – 87%	B
73 – 80%	BC

Percentage	Grade
60 – 73%	C
55 – 60%	D
0 – 55%	F

Approximate schedule (See the course web site for the definitive schedule.)

Date	Topic	Reading
Jan 19	Overview	Ch 1
Jan 24	Meiosis and recombination	
Jan 26	**No lecture**	
Jan 31	Hidden Markov models (HMMs) for QTL mapping	App D
Feb 2	Data diagnostics	Ch 3
Feb 7	**No lecture**	
Feb 9	Genetic map construction	tech report
Feb 14	Interval mapping	Sec 4.1-4.2
Feb 16	Significance thresholds; the X chromosome	Sec 4.3-4.4
Feb 21	Interval estimates of QTL location	Sec 4.5
Feb 23	Non-normal traits I	Ch 5
Feb 28	Non-normal traits II	
Mar 2	Experimental design and power	Ch 6
Mar 7	Covariates	Ch 7
Mar 9	Two-dimensional, two-QTL genome scans	Ch 8
Mar 14	**No lecture** (Spring Break)	
Mar 16	**No lecture** (Spring Break)	
Mar 21	Mapping multiple QTL I	Ch 9
Mar 23	Mapping multiple QTL II	
Mar 28	Bayesian methods I	
Mar 30	Bayesian methods II	
Apr 4	Recombinant inbred lines	
Apr 6	Panels of consomics or congenics	
Apr 11	Advanced intercross lines	
Apr 13	Heterogeneous stock and the Collaborative Cross	
Apr 18	Outbred crosses	
Apr 20	Combining crosses	
Apr 25	eQTL analysis I	
Apr 27	eQTL analysis II	
May 2	Research topics	
May 4	Summary	